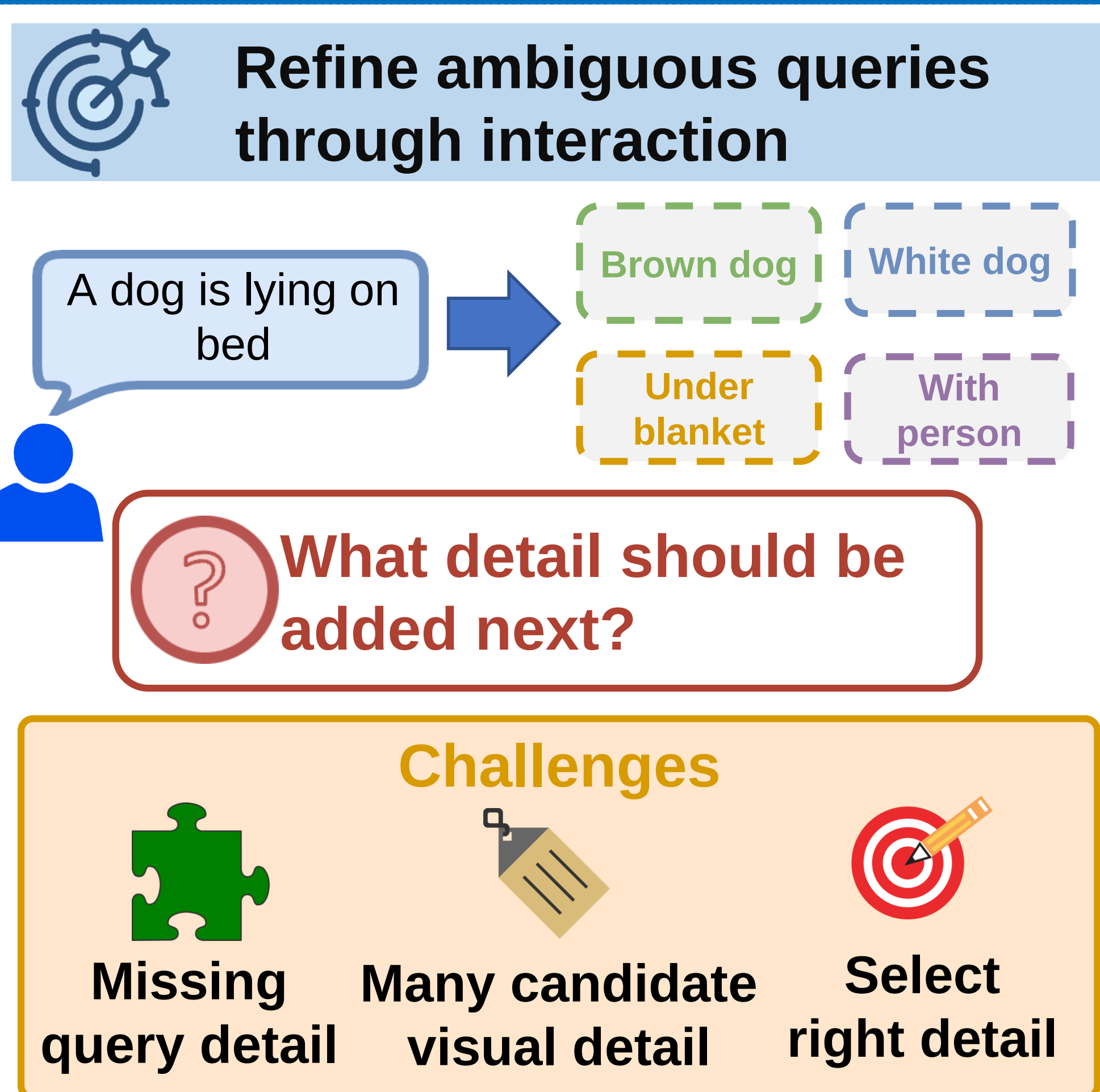


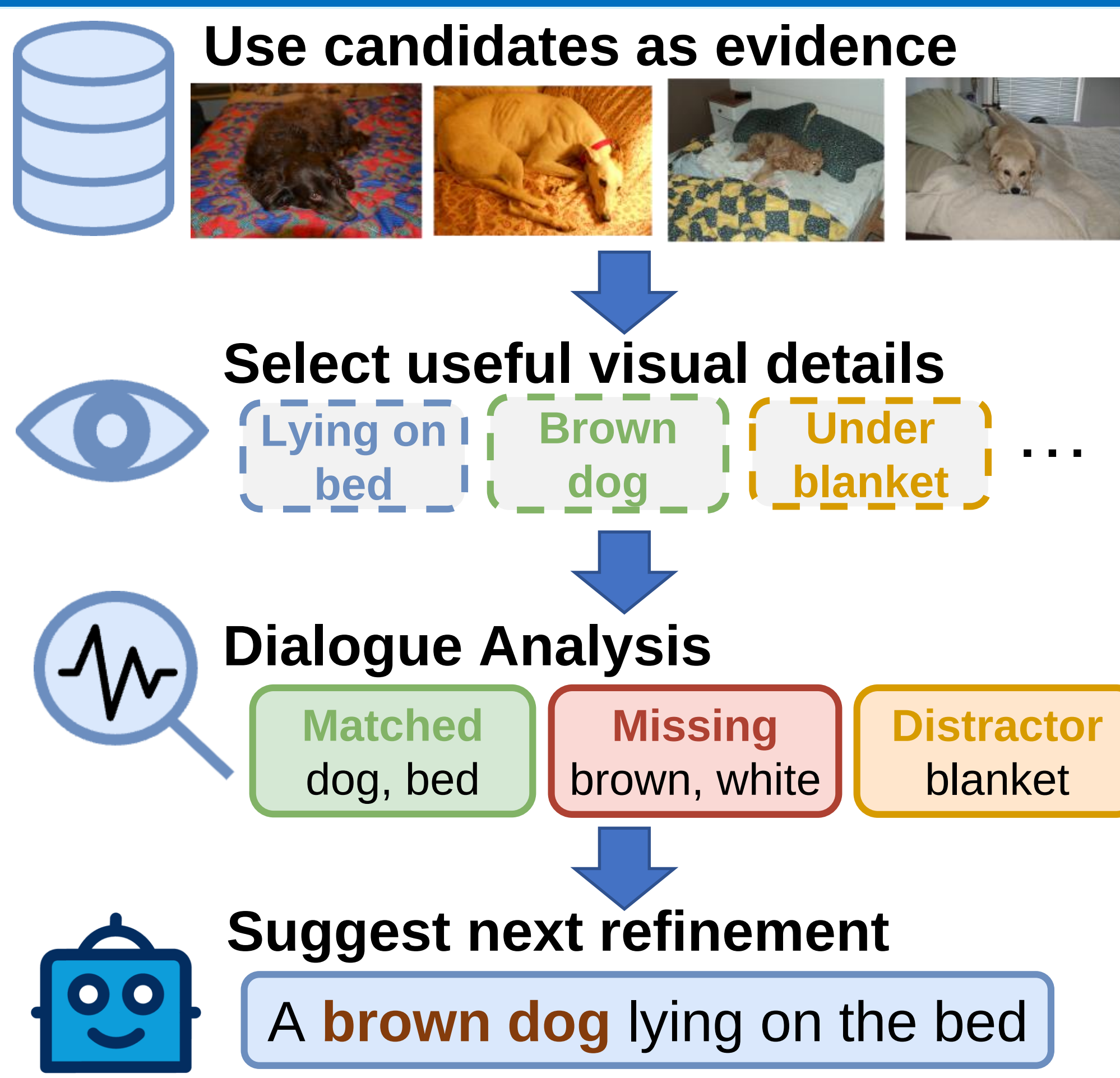
1 Traditional Retrieval



2 Goal & Technical Challenges

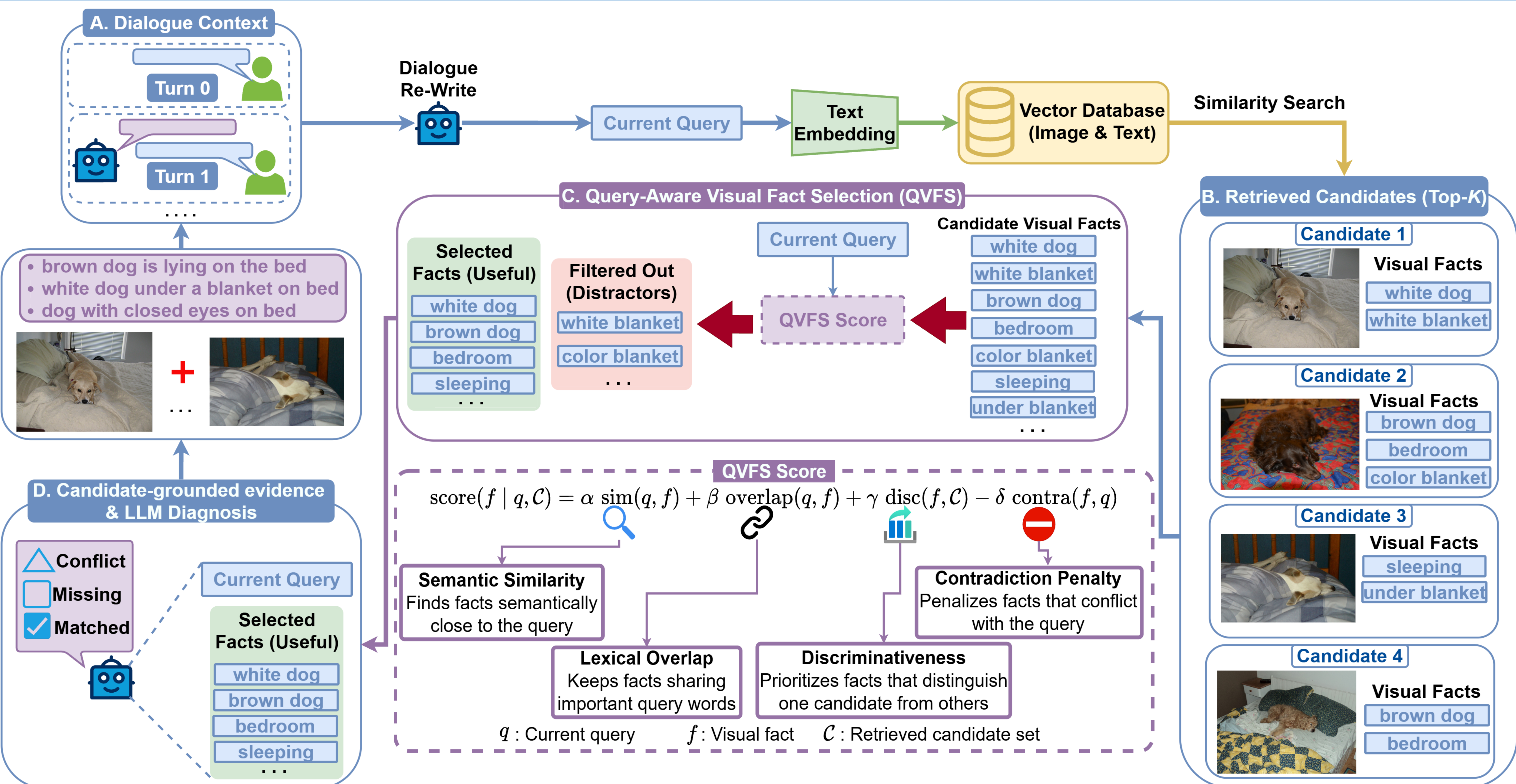


3 Proposed Idea



4 Proposed Framework: RAIR-VF

RAIR-VF: Retrieval candidate images, select compact visual facts, diagnose matched / missing / conflicting constraints, and suggest refined queries.



5 Experiments & Results

Dataset: Visual Dialog (VisDial) [1], 122k images

Metric: Hit@K

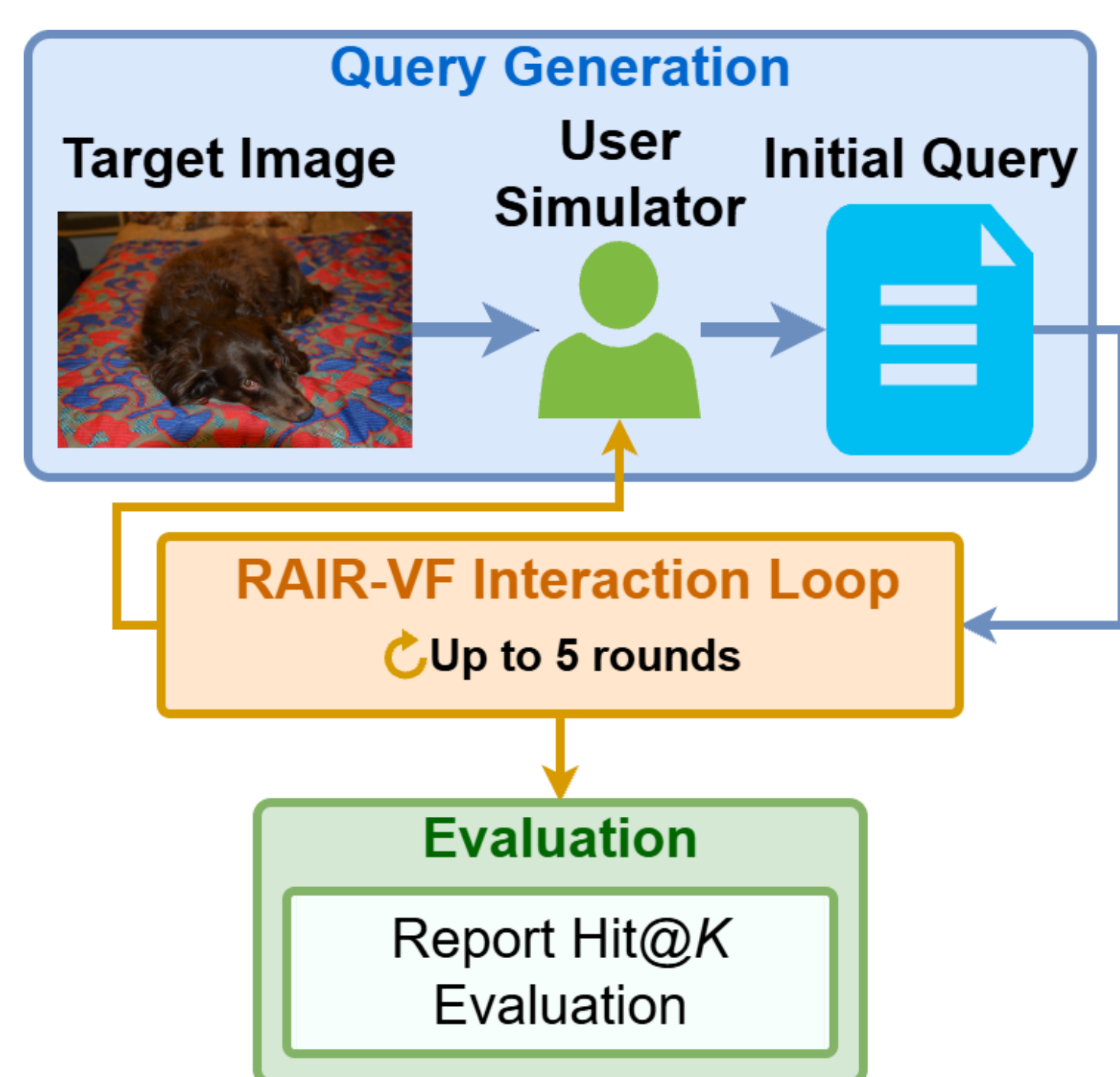
Local LLM: Gemma3-12B [2]

User simulator: Qwen2.5-VL [3]

Database: PostgreSQL,

Visual encoder: CLIP-ViT [4]

Turn	Hit@10	Hit@20	Hit@50
0	0.4444	0.5456	0.6769
1	0.4788	0.5794	0.7125
2	0.4913	0.6394	0.7856
3	0.5320	0.6931	0.8212
4	0.5969	0.7450	0.8531
5	0.6350	0.7756	0.8731



6 Conclusion & Future Works

Conclusion

- **Candidate-grounded visual facts:** Support query refinement.
- **QVFS:** Provides compact and relevant evidence.
- **LLM diagnosis:** Helps clarify missing or conflicting constraints.
- **Multi-turn interaction:** Improves best-so-far retrieval performance.

Future Works:

- Deploy an **interactive user application** on a Web application platform.
- Conduct **human-user studies** to evaluate the usefulness and interpretability of refinement suggestions.